

MCCREARY, MB, Canada

WMO: 711490

Lat: 50.7119N

Lon: 99.5300W

Elev: 351

StdP: 97.18

Time Zone: -6.00 (NAC)

Period: 97-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -29.5	(c) -27.0	(d) -33.5	(e) 0.2	(f) -28.8	(g) -31.1	(h) 0.2	(i) -26.4	(j) 12.2	(k) 0.1	(l) 10.8	(m) -2.7	(n) 2.3	(o) 320	(p) 0.617

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	29.8	20.4	27.9	19.5	26.3	18.5	22.4	27.1	21.1	26.0	19.9	24.7	3.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.9	16.2	25.2	19.4	14.8	23.8	18.1	13.6	22.7	67.8	27.2	62.9	26.1	58.7	24.7	28.7

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.5	8.3	7.3	-33.7	33.5	2.5	2.0	-35.5	34.9	-37.0	36.1	-38.4	37.2	-40.2	38.6		
			-33.8	25.0	2.5	1.9	-35.6	26.4	-37.1	27.6	-38.5	28.6	-40.3	30.1		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	3.5	-13.6	-12.2	-5.3	4.0	10.6	16.0	19.5	18.3	13.2	5.3	-3.4	-11.5		
	DBStd	13.30	8.49	7.71	7.79	6.37	5.12	3.84	3.33	3.53	4.60	5.13	6.89	8.04		
	HDD10.0	3333	733	622	473	196	55	3	0	0	21	163	403	665		
	HDD18.3	5553	991	856	731	429	245	88	27	44	164	403	651	923		
	CDD10.0	961	0	0	0	17	73	183	294	258	116	18	2	0		
	CDD18.3	141	0	0	0	1	5	18	63	44	9	1	0	0		
	CDH23.3	1273	0	0	0	5	71	160	541	398	93	7	0	0		
	CDH26.7	302	0	0	0	0	13	29	132	110	17	1	0	0		
(14) Wind	WSAvg	3.4	3.5	3.4	3.6	3.8	3.7	3.0	3.0	3.1	3.3	3.7	3.6	3.4		
(15) Precipitation	PrecAvg	555	20	16	34	33	73	95	77	64	53	40	22	22		
	PrecMax	679	50	53	111	62	186	165	176	163	98	130	104	57		
	PrecMin	406	3	1	1	8	20	36	26	11	10	6	0	3		
	PrecStd	74	12	10	25	17	37	36	39	37	25	28	23	12		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.9	7.4	14.6	23.6	28.1	29.9	32.0	32.5	28.9	23.6	16.3	8.0		
		MCWB	3.6	2.9	7.6	12.6	15.7	20.2	22.8	21.4	18.3	14.1	9.3	3.3		
	2%	DB	4.5	4.2	10.9	20.1	25.3	26.9	29.8	29.6	26.0	19.0	11.8	5.0		
		MCWB	1.6	1.0	5.9	10.5	14.0	18.6	21.3	20.1	18.1	11.5	6.3	1.7		
	5%	DB	1.9	1.8	8.0	17.3	22.6	25.1	28.0	27.4	23.5	16.1	8.5	2.3		
		MCWB	-0.5	-0.1	4.0	9.3	13.0	17.4	20.3	19.5	16.4	10.0	4.3	-0.2		
	10%	DB	-1.0	-0.5	5.2	14.3	19.7	23.3	26.4	25.4	21.0	13.5	5.8	-0.5		
		MCWB	-2.5	-1.8	2.5	7.8	11.8	16.6	19.4	18.5	14.8	8.7	2.7	-2.3		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	3.6	3.2	8.6	13.5	17.9	23.1	25.2	23.6	21.2	14.5	9.4	3.5		
		MCDB	7.2	6.1	12.6	20.7	24.6	26.9	29.3	29.2	25.4	21.4	16.1	7.3		
	2%	WB	1.8	1.5	6.2	11.3	15.6	20.4	22.9	21.8	19.2	12.3	6.8	1.8		
		MCDB	4.4	3.7	10.4	18.6	21.5	24.6	27.5	27.1	23.8	17.4	11.1	4.8		
	5%	WB	-0.3	0.0	4.2	9.7	14.1	18.7	21.5	20.6	17.3	10.6	4.7	-0.1		
		MCDB	1.6	1.8	7.5	16.1	20.3	22.9	26.3	25.8	21.8	14.8	8.0	2.2		
	10%	WB	-2.5	-2.0	2.5	8.2	12.7	17.5	20.3	19.4	15.7	9.1	2.9	-2.1		
		MCDB	-1.0	-0.4	5.2	13.9	18.3	21.9	25.1	24.1	20.0	13.1	5.8	-0.6		
(35) Mean Daily Temperature Range	5% DB	MDBR	9.2	9.4	9.7	11.0	11.4	10.1	10.7	11.0	10.5	8.9	7.9	8.6		
		MCDBR	11.3	10.9	11.8	15.4	15.5	12.7	12.7	13.9	14.4	13.7	11.5	10.9		
	5% WB	MCWBR	9.0	8.4	7.9	8.5	7.7	6.8	6.7	7.3	7.9	7.6	7.3	8.4		
		MCWBR	10.9	10.2	11.3	14.4	13.0	11.3	11.9	12.1	13.0	12.5	11.1	10.7		
(40) Clear-Sky Solar Irradiance	taub		0.253	0.259	0.297	0.342	0.357	0.370	0.385	0.378	0.342	0.295	0.263	0.237		
	taud		2.288	2.298	2.316	2.369	2.391	2.379	2.359	2.375	2.459	2.553	2.457	2.353		
	Ebn at Noon		788	888	909	901	899	886	865	854	852	845	785	765		
	Edh at Noon		65	85	102	110	113	115	116	108	89	67	56	52		
(44) All-Sky Solar Radiation	RadAvg		1.31	2.51	3.91	5.09	5.55	5.84	6.17	5.12	3.68	2.15	1.16	0.94		
	RadStd		0.11	0.15	0.29	0.35	0.50	0.41	0.27	0.39	0.29	0.28	0.16	0.09		

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
Station Only	N/A	N/A	N/A	N/A	-1.04	-1.52	N/A	N/A	N/A	N/A
Regional (22 neighbors)	N/A	N/A	N/A	N/A	-0.50	-0.67	N/A	N/A	N/A	N/A

Nomenclature: See separate page